Daffodil International University
Department of Computer Science and Engineering

SE 313 — Software Quality Assurance & Testing Lab
Lab Evaluation

Time: 40 Minutes	Total Marks: 25	Viva: 5 Marks (included)

Note: 5 marks are reserved for viva voce. Be ready to explain your design choices, justify partition boundaries, and defend your test case selection.

Instructions:

- Answer all questions. Partial credit is given for correct logic even with minor syntax errors.
- Clearly label each partition (valid/invalid) and each boundary value you choose.
- Write clean, compilable Java code where asked.
- No discussion with classmates during the evaluation.

Question 1 — Equivalence Partitioning & Boundary Value Analysis [8 Marks]

An online course registration system validates two fields:

- **Age:** must be between 18 and 60, inclusive.
- **Password length:** must be between 6 and 12 characters, inclusive.

a) Identify all equivalence partitions (valid and invalid) for the age field. Pick one representative test value from each partition and justify your choice. **(2 marks)**

b) List all boundary values that must be tested for the password length requirement. Include values just inside and just outside each boundary, and state which partition each belongs to. **(2 marks)**

c) In 2–3 sentences, explain why boundary values are more likely to reveal defects than values chosen from the middle of a valid range. **(1 mark)**

d) Write Java methods **isValidAge(int age)** and **isValidPassword(String password)** that implement these checks. Write a **main** method that tests each against the boundary values you identified in (b) and prints the length and validity for each password. Include at least one test from each partition for age. **(3 marks)**

Question 2 — Integrated Registration Scenario [8 Marks]

Using the four requirements below, design and implement a complete registration validator:

- Age: 18–60 inclusive
- Password length: 6–12 inclusive
- Course: must be exactly Java, Python, or C++ (case-insensitive)
- Scholarship applied: true or false

a) Design **three test cases** that combine all four fields: one that should be accepted, and two that should be rejected for *different* reasons. Present them in a table with columns: Age, Password, Course, Scholarship, Expected Result, Reason. **(3 marks)**

b) Implement a Java method:

```
register(int age, String password, String course, boolean scholarshipApplied)
```

that validates all four fields and prints whether registration is accepted or rejected, naming any field(s) that caused rejection. Run it against your three test cases from (a). **(3 marks)**

c) The course field accepts exactly three discrete values. Write a Java method **isValidCourse(String course)** that handles case-insensitivity and null input. Also explain in one sentence why a boolean field requires only two test cases for full partition coverage. **(2 marks)**

Question 3 — Conceptual Questions [4 Marks]

a) Explain the difference between a **closed interval** [10, 20] and an **open interval** (10, 20) in the context of boundary value testing. For the value 20, state whether it belongs to each interval and explain why this distinction matters when writing boundary test cases. **(2 marks)**

b) Explain in your own words what equivalence partitioning achieves and when you would choose boundary value analysis *instead of* (or in addition to) equivalence partitioning. **(2 marks)**

Viva Questions — For Examiner Use [5 Marks]

- Why do we test boundary values rather than only the middle of a valid range?
- If the age requirement changes from 18–60 to 18–65, which test cases would you need to update and why?
- What is the risk of testing only one value from an equivalence class instead of a boundary value?
- How would you extend the Question 2 validator if a fifth field (email format) were added?
- What is the difference between a valid partition and an invalid partition? Give an example from today's questions.

— End of Evaluation —